A PROJECTIVE IDEAL THAT IS NOT LOCALLY FREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The projectivity lemma's own proof (idempotent normalization, Hom colimit, Mittag-Leffler) is delivered in L48286 and is not repeated here.

Editorial note. The label lemma-ideal-projective-not-locally-free names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Delivered card L48286 (same unit pool): an ideal generated by a countable collection of idempotents is projective as an R-module (author lemma-ideal-generated-by-idempotents-projective, source lines 2661-2683, with its complete Mittag-Leffler proof)
- Context span 2685-2694: if P subset Q is an inclusion of R-modules with Q finite and P locally free, and Q is generated by N elements, then P is finite locally free with free parts of rank at most N (Algebra, map-cannot-be-injective), and P is a finite R-module (Algebra, finite-projective)
- Context span 2696-2705: combining these, a non-finitely-generated ideal generated by a countable collection of idempotents is projective but not locally free; explicit example R = product over N of F_2 with I generated by the idempotents e_n = (1, ..., 1, 0, ...) with n ones

Source reference. Original definition 00NW: sources/stacks-chow/tags.txt, source lines 874–874. Prerequisite pointer invoked at 2719; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Prerequisite pointer invoked at 2725; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05WI: sources/stacks-chow/tags.txt, source lines 7587–7587. Prerequisite pointer invoked at 2722; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Suppose that $P \subset Q$ is an inclusion of R-modules with Q a finite R-module and P locally free, see Algebra, Definition 00NW. Suppose that Q can be generated by N elements as an R-module. Then it follows from Algebra, Lemma 05WI that P

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is finite locally free (with the free parts having rank at most N). And in this case P is a finite R-module, see Algebra, Lemma 00NX.

Combining this with the above we see that a non-finitely-generated ideal which is generated by a countable collection of idempotents is projective but not locally free. An explicit example is $R = \prod_{n \in \mathbf{N}} \mathbf{F}_2$ and I the ideal generated by the idempotents

$$e_n = (1, 1, \ldots, 1, 0, \ldots)$$

where the sequence of 1's has length n.

Lemma 0.1. *There exists a ring R and an ideal I such that I is projective as an R-module but not locally free as an R-module.*

Proof. See above. □